

4.2 $\varepsilon - \delta$ definition of continuity

Before giving the formal definition of continuity, let us first understand its basic idea. Suppose we want the value of a function to be close to a desired output $f(a)$. A natural question is:

How close should the input x be to a so that the output $f(x)$ is as close as we want to $f(a)$?

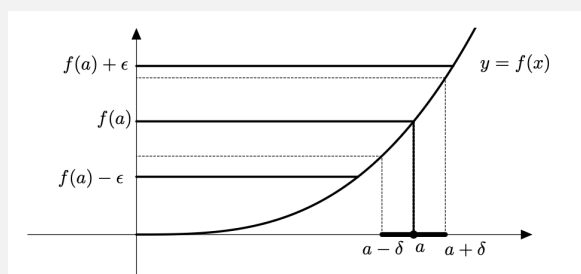
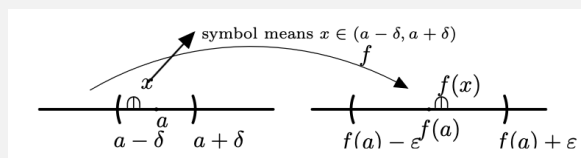
The answer depends on how much error we are willing to tolerate in the output. If we specify a tolerance $\varepsilon > 0$ for the output, continuity guarantees that there exists a tolerance $\delta > 0$ for the input such that whenever x is within δ of a , the corresponding value $f(x)$ is within ε of $f(a)$.

This simple idea forms the basis of the classical ε - δ definition of continuity.

Definition 28. Let $J \subseteq \mathbb{R}$, let $f : J \rightarrow \mathbb{R}$ be a function, and let $a \in J$. We say that f is continuous at a if for every $\varepsilon > 0$, there exists $\delta > 0$ such that

$$x \in J \text{ and } |x - a| < \delta \implies |f(x) - f(a)| < \varepsilon.$$

We say that f is continuous on J if it is continuous at every point of J .



Theorem 29. The sequential definition of continuity (Definition 27) and the ε - δ definition (Definition 28) are equivalent.

Proof. We prove both implications.

(ε - δ continuity $\Rightarrow$ sequential continuity). Suppose f is continuous at a in the sense of Definition 28. Let (x_n) be a sequence in J such that $x_n \rightarrow a$. Given $\varepsilon > 0$, there exists $\delta > 0$ such that

$$x \in J, \quad |x - a| < \delta \quad \Longrightarrow \quad |f(x) - f(a)| < \varepsilon.$$

Since $x_n \rightarrow a$, there exists $N \in \mathbb{N}$ such that

$$n \geq N \quad \Longrightarrow \quad |x_n - a| < \delta.$$

Hence

$$n \geq N \quad \Longrightarrow \quad |f(x_n) - f(a)| < \varepsilon.$$

Thus $f(x_n) \rightarrow f(a)$, so f is continuous at a in the sense of Definition 27.

(Sequential continuity $\Rightarrow$ ε - δ continuity). Suppose f is continuous at a in the sense of Definition 27. Assume, to the contrary, that the ε - δ condition fails. Then there exists $\varepsilon_0 > 0$ such that for every $\delta > 0$, there is $x \in J$ satisfying

$$|x - a| < \delta, \quad |f(x) - f(a)| \geq \varepsilon_0.$$

For each $n \in \mathbb{N}$, choose $x_n \in J$ such that

$$|x_n - a| < \frac{1}{n}, \quad |f(x_n) - f(a)| \geq \varepsilon_0.$$

Then $x_n \rightarrow a$. By sequential continuity, $f(x_n) \rightarrow f(a)$. Hence there exists $N \in \mathbb{N}$ such that

$$n \geq N \quad \Longrightarrow \quad |f(x_n) - f(a)| < \varepsilon_0,$$

contradicting $|f(x_n) - f(a)| \geq \varepsilon_0$ for every n .

Therefore, the ε - δ condition holds, and the two definitions are equivalent. $\square$

Example 67. Show that the identity function

$$f : \mathbb{R} \rightarrow \mathbb{R}, \quad f(x) = x,$$

is continuous on $\mathbb{R}$.

Solution. Let $a \in \mathbb{R}$ and let $\varepsilon > 0$ be given. We have

$$|f(x) - f(a)| = |x - a|.$$

Thus, it is natural to choose

$$\delta = \varepsilon.$$

Then, whenever $|x - a| < \delta$, we obtain

$$|f(x) - f(a)| = |x - a| < \delta = \varepsilon.$$

Hence, f is continuous at a . Since a was arbitrary, f is continuous on $\mathbb{R}$.

Example 68. Suppose $R > 0$. Show that the function

$$f : [-R, R] \rightarrow \mathbb{R}, \quad f(x) = x^2,$$

is continuous.

Solution. Let $a \in [-R, R]$ and let $\varepsilon > 0$ be given. Observe that

$$|f(x) - f(a)| = |x^2 - a^2| = |x + a||x - a| \leq (|x| + |a|)|x - a| \leq 2R|x - a|.$$

Choose

$$\delta = \frac{\varepsilon}{2R}.$$

Then, whenever $|x - a| < \delta$, we have

$$|f(x) - f(a)| \leq 2R|x - a| < 2R\delta = \varepsilon.$$

Hence, f is continuous on $[-R, R]$.

Exercise 18. Show that the function $f : \mathbb{R} \rightarrow \mathbb{R}$, defined by $f(x) = x^2$, is continuous on $\mathbb{R}$ using the ε - δ definition. Hint. Observe that $|x^2 - a^2| = |x - a||x + a|$. To estimate $|x + a|$, first assume $\delta \leq 1$. Then, whenever $|x - a| < \delta$, we have

$$|x| \leq |x - a| + |a| < 1 + |a|,$$

and hence

$$|x + a| \leq |x| + |a| < 1 + 2|a|.$$

Now, choose $\delta = \min \left\{ 1, \frac{\varepsilon}{1 + 2|a|} \right\}$.

Theorem 30. Let $J \subseteq \mathbb{R}$ and let $f : J \rightarrow \mathbb{R}$ be continuous at $c \in J$. If $f(c) \neq 0$, then there exists $\delta > 0$ such that $|f(x)| > \frac{|f(c)|}{2}$ for all $x \in (c - \delta, c + \delta) \cap J$.

In particular, if $f(c) > 0$, then there exists $\delta > 0$ such that $f(x) > \frac{f(c)}{2}$ for all $x \in (c - \delta, c + \delta) \cap J$.

Proof. Choose $\varepsilon = \frac{|f(c)|}{2}$. Since f is continuous at c , there exists $\delta > 0$ such that $|f(x) - f(c)| < \varepsilon$ whenever $x \in (c - \delta, c + \delta) \cap J$. Hence,

$$|f(x)| \geq |f(c)| - |f(x) - f(c)| > |f(c)| - \frac{|f(c)|}{2} = \frac{|f(c)|}{2}.$$

If $f(c) > 0$, then $|f(x) - f(c)| < \frac{f(c)}{2}$ implies $f(x) > \frac{f(c)}{2}$. □

4.3 Properties of continuous functions

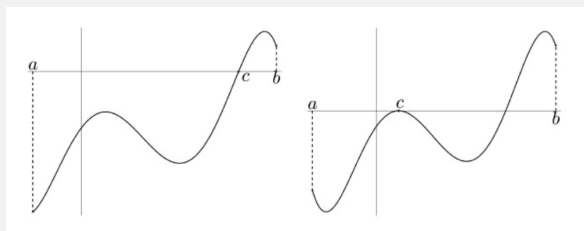
Suppose $f : [a, b] \rightarrow \mathbb{R}$ is a continuous function such that $f(a)$ and $f(b)$ have opposite signs. Draw the graphs of a few such functions, considering both the cases $f(a) > 0 > f(b)$ and $f(a) < 0 < f(b)$.

What do you observe? Does the graph always intersect the x -axis? If so, can you explain why this must happen? In other words, must there exist a point $c \in (a, b)$ such that $f(c) = 0$?

The answer is affirmative, and it is one of the most important consequences of continuity. This result is known as the *Intermediate Value Theorem*.

Theorem 31 (Intermediate Value Theorem). *Let $f : [a, b] \rightarrow \mathbb{R}$ be a continuous function such that $f(a) < 0 < f(b)$. Then there exists a point $c \in (a, b)$ satisfying $f(c) = 0$.*

Sketch of the proof. The idea is illustrated in the following figure.



We look for the *first* point where the graph meets the x -axis. To do this, consider the set

$$E = \{x \in [a, b] : f(y) \leq 0 \text{ for every } y \in [a, x]\}.$$

Clearly, E is nonempty (since $a \in E$) and bounded above by b . By the Completeness Property of $\mathbb{R}$, the set E has a least upper bound; let

$$c = \sup E.$$

Using the continuity of f , one shows that neither $f(c) < 0$ nor $f(c) > 0$ is possible. Therefore,

$$f(c) = 0.$$

Hence, there exists $c \in (a, b)$ such that $f(c) = 0$. □

Theorem 32 (Intermediate Value Theorem (Standard Version)). *Let $g : [a, b] \rightarrow \mathbb{R}$ be a continuous function, and let λ be a real number between $g(a)$ and $g(b)$ that is $\min\{g(a), g(b)\} < \lambda < \max\{g(a), g(b)\}$. Then there exists a point $c \in (a, b)$ such that*

$$g(c) = \lambda.$$

Proof. Without loss of generality, assume that $g(a) < \lambda < g(b)$. Define

$$f(x) = g(x) - \lambda.$$

Since g is continuous on $[a, b]$, so is f . Moreover,

$$f(a) = g(a) - \lambda < 0 \quad \text{and} \quad f(b) = g(b) - \lambda > 0.$$

Hence, by the Intermediate Value Theorem [31](#), there exists $c \in (a, b)$ such that

$$f(c) = 0.$$

Therefore,

$$g(c) = \lambda.$$

□

Exercise 19. Show that if $f : \mathbb{R} \rightarrow \mathbb{R}$ is a continuous function whose range is contained in $\mathbb{Z}$, then f is constant.